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On-Device Low-Frequency Mechanical Fault Detection of Induction Motors using MEMS Sensors and Edge AI

Vedanta Banerjee, Debjyoti Chowdhury, Madhurima Chattopadhyay
Department of Applied Electronics and Instrumentation Engineering
Heritage Institute of Technology, Kolkata, India
vedantabanerjee01@gmail.com, {debjyoti.chowdhury, madhurima.chattopadhyay}@heritageit.edu

***Abstract*—This paper presents a TinyML system for end-to-end, on-device vibration-based mechanical fault detection in single-phase induction motors. The sensing node integrates a low-cost MEMS inertial sensor (LSM6DSO) with an ESP32-C3 microcontroller. The proposed firmware implements orientation-invariant preprocessing, contiguous leakage-safe segmentation, and int8 quantized inference of a compact multi-channel 1D Convolutional Neural Network (CNN). Window-level outputs are aggregated using a confidence-weighted majority voting rule to provide robust diagnoses across four states: motorOFF, motorON, motorON_NoFan, and motorON_BadFan. By employing strictly contiguous file splits to prevent temporal data leakage, we report a window-level macro F1-score of 0.98 and a recording-level macro F1-score of 0.95 on held-out, position-specific data. The quantized model requires only 80 KB of Flash and 120 KB of RAM, with an inference latency of 82 ms. This demonstrates the feasibility of deploying sophisticated diagnostic AI on low-power edge devices for real-time industrial condition monitoring of induction motors.**

***Index Terms*—Condition monitoring, convolutional neural networks (CNN), edge AI, induction motors, microcontrollers, predictive maintenance, TinyML, vibration analysis.**

I. Introduction

Induction motors (IMs) are the industrial backbone, powering critical infrastructure from HVAC systems to process conveyors [1]. Given their continuous operation in harsh environments, reliable performance is a primary driver of plant availability and energy efficiency. Even marginal improvements in early fault detection can significantly reduce downtime, repair costs, and safety risks [2]. While industrial monitoring often prioritizes bearing failures, this study focuses on low-frequency mechanical anomalies, specifically fan unbalance and removal. These faults represent critical failure modes that lead to motor overheating and accelerated insulation degradation.

A. Induction Motor Fault Modalities

IM faults are broadly categorized as **electrical** (e.g., winding anomalies, broken bars) or **mechanical** (e.g., bearing defects, misalignment, and fan damage). Electrical faults are typically diagnosed through current-signature analysis and time-frequency processing of stator currents [3], [4]. Conversely, mechanical faults manifest as localized vibration signatures requiring accelerometer-based envelope or spectral analysis [5]. Combining these modalities often enhances diagnostic breadth and reliability across varied operating conditions [6].

B. Vibration-Based Diagnostics and Edge Computing

Vibration sensing captures the mechanical energy imparted to motor housings, making it ideal for detecting the impulsive or modulated patterns characteristic of shaft misalignment and fan-related anomalies. While classical time-domain descriptors (RMS, kurtosis) and frequency-domain peaks offer interpretability, advanced time-frequency techniques like STFT or wavelets are essential for non-stationary conditions. Recent advancements in MEMS accelerometers facilitate cost-effective, distributed monitoring networks when integrated with energy-efficient edge nodes [7], enabling sophisticated diagnostics without reliance on cloud-based processing.

II. Related Work

A. Review of Literature and Prior Art

Induction motor fault diagnosis has evolved from canonical signal processing to sophisticated machine learning. Established surveys emphasize that robust condition monitoring requires specialized preprocessing for non-stationary signals and careful consideration of mechanical coupling effects [1]. Classical methodologies typically utilize engineered time- and frequency-domain features paired with SVMs or tree-based classifiers to distinguish mechanical faults in controlled environments [5]. Comparative studies indicate that while current-signature analysis is effective for electromechanical interactions, vibration-based sensing remains more sensitive to purely mechanical anomalies [6].

Recent advancements favor end-to-end deep learning, where CNNs trained on time-frequency representations achieve high diagnostic accuracy, particularly on standard bearing datasets [8], [9]. These successes motivate the shift toward TinyML, where models are quantized for microcontroller deployment. However, existing literature reveals critical methodological gaps: (i) a lack of reproducible metrics for full TinyML stacks (Flash, RAM, and latency), (ii) insufficient exploration of mounting-position variability and orientation-invariant preprocessing, (iii) a lack of real-time synchronization between hardware acquisition and inference, and (iv) prevalent data

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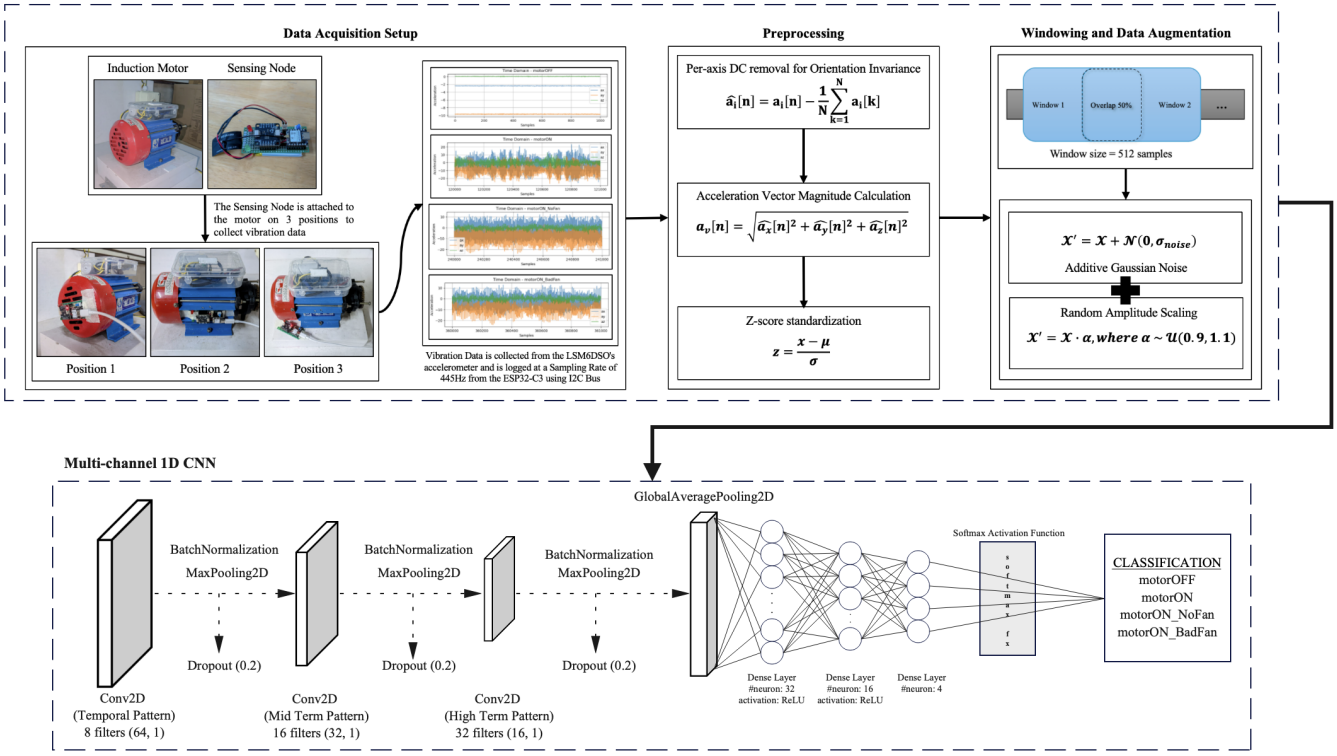


Fig. 1: Proposed end-to-end TinyML system block diagram for on-device fault classification.

leakage caused by random shuffling of overlapping temporal windows.

This work addresses these gaps by implementing a **leakage-safe, orientation-invariant pipeline** on an ESP32-C3 microcontroller. Unlike studies focusing solely on bearing datasets, we utilize a multi-channel 1D CNN to specifically target **low-frequency fan-related mechanical faults**, ensuring strictly contiguous data splits for rigorous evaluation.

III. System Overview

The end-to-end architecture is illustrated in the block diagram in Fig. 1, which shows the flow from raw data acquisition via the LSM6DSO to on-device confidence-weighted inference. The physical mounting logic is further detailed in Fig. 3, where the sensor is placed at three distinct points to capture position-dependent vibrational coupling.

A. Hardware Architecture and Sensor Placement

The sensing node (Fig. 2) is designed for high-fidelity vibration acquisition within a constrained power budget. It integrates an LSM6DSO 6-DoF IMU with a Seeed Studio XIAO ESP32-C3, featuring a 160 MHz RISC-V core and 400 KB SRAM. The tri-axial accelerometer is configured to a full-scale range of $\pm 4g$, providing sufficient dynamic range for industrial motor vibrations. To optimize battery longevity while ensuring signal integrity for low-frequency faults, data is sampled at $f_s = 445$ Hz. This configuration directly aligns with the IMU's internal Output Data Rate (ODR) to prevent aliasing and satisfies the Nyquist criterion for mechanical anomalies manifesting below 220 Hz.

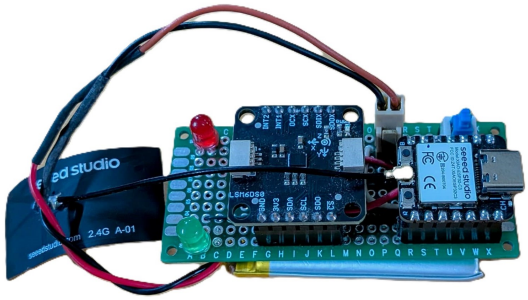


Fig. 2: Compact edge sensor node



Position 1 Position 2 Position 3

Fig. 3: Sensor placement positions

As illustrated in Fig. 3, three specific mounting positions

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were selected to characterize the vibration transduction across the motor frame:

- **Position 1 (Drive End):** Located near the load coupling to capture shaft-related vibrations and torque-induced oscillations.
- **Position 2 (Central):** Positioned at the mid-housing to monitor core electromagnetic vibrations and overall stator resonance.
- **Position 3 (Non-Drive End):** Placed adjacent to the cooling fan. This position is critical for our study, as it provides the highest signal-to-noise ratio for detecting fan unbalance and aerodynamic turbulence.

Collecting data across these varied locations allows the model to learn position-invariant features, ensuring that the diagnostic AI remains robust regardless of where a technician mounts the device on the motor housing.

B. Dataset layout and sample counts

The collected data is organized into datasets, which are then organized into two directories: `datasets/train/` and `datasets/test/`. File-level organization and counts are summarized in Table I.

TABLE I: Dataset files and raw sample counts

File / Set (`datasets/`)	Samples per file	Label / Notes
`train/motorOFF.csv`	**120000***	class 0
`train/motorON.csv`	**120000***	class 1
`train/motorON_NoFan.csv`	**120000***	class 2
`train/motorON_BadFan.csv`	**120000***	class 3
Total (train dir)	**480000**	**balanced across all four classes**
`test/motorOFF_P1.csv`	**10000**	held-out position P1
`test/motorOFF_P2.csv`	**10000**	held-out position P2
`test/motorOFF_P3.csv`	**10000**	held-out position P3
⋮	⋮	⋮
`test/motorON_BadFan_P3.csv`	**10000**	held-out position P3
Total (test dir)	**120000**	**12 files × 10,000**

*Each class file in `datasets/train` aggregates **40,000** samples from each position - P1, P2 and P3, amounting to **120,000** samples for each class.

IV. Preprocessing and Windowing

A. Signal Transformation and Orientation Invariance

To ensure the model is robust to varied sensor mounting angles, a two-stage preprocessing pipeline is implemented.

First, for each contiguous recording segment of length N, we perform **per-axis DC removal**. Let $a_i[n]$ represent the raw acceleration for axis $i \in \{x, y, z\}$. The zero-mean signal $\hat{a}_i[n]$ is computed as:

$$\hat{a}_i[n] = a_i[n] - \frac{1}{N}\sum_{k=1}^{N} a_i[k] \tag{1}$$

This step effectively eliminates the static gravity component $1g$, ensuring that the vibration signatures are independent of the sensor's spatial orientation.

Next, we derive the **Acceleration Vector Magnitude** (Resultant Acceleration) $a_v[n]$ to capture the total vibrational energy:

$$a_v[n] = \sqrt{\hat{a}_x[n]^2 + \hat{a}_y[n]^2 + \hat{a}_z[n]^2} \tag{2}$$

Finally, all four channels are subjected to **Z-score standardization**. Each sample x is transformed using the global training mean μ and standard deviation σ as:

$$z = \frac{x - \mu}{\sigma} \tag{3}$$

B. Windowing and Data Augmentation

The continuous data stream is partitioned into temporal windows using a sliding window approach. We define a window length $L = 512$ samples and a stride $S = 256$, resulting in a 50% overlap to preserve transient features at segment boundaries. Each input instance $\mathcal{X} \in \mathbb{R}^{512\times 4}$ represents the four channels $(\hat{a}_x, \hat{a}_y, \hat{a}_z, a_v)$.

During the training phase, on-the-fly augmentations are applied to improve model generalization:

- **Additive Gaussian Noise:** $\mathcal{X}' = \mathcal{X} + \mathcal{N}(0, \sigma_{noise})$, where σ_{noise} is 1% of the signal's standard deviation.
- **Random Amplitude Scaling:** $\mathcal{X}' = \mathcal{X} \cdot \alpha$, where $\alpha \sim \mathcal{U}(0.9, 1.1)$.

Augmentations were disabled during validation and held-out testing to ensure a deterministic assessment of base model performance.

V. Model Design and Training

A. Architecture

The model is a compact **multi-channel 1D CNN** architecture (implemented using Conv2D layers with $N \times 1$ kernels) optimized for low parameter count and quantization friendliness. It utilizes elongated temporal kernels to capture waveform envelopes, small channel counts, batch normalization, and dropout for regularization. The feature extractor is followed by global average pooling and a small dense head. Total trainable parameters are approximately $\mathbf{15k}$. The layer configurations in Table II correspond to the input window size of $L = 512$.

B. Training recipe

Optimization uses Adam (initial LR $= 1 \times 10^{-3}$), batch size 32, sparse categorical cross-entropy, ReduceLROnPlateau (patience 8), and EarlyStopping (patience 12). Representative calibration windows ($\mathbf{1k}$–$\mathbf{5k}$, uniformly sampled across mount positions and classes) are used for post-training integer quantization (PTQ).

VI. On-Device Integration and Resource Profile

A. Quantization and TFLM integration

We apply full integer PTQ (int8) and integrate the '.tflite' model into TensorFlow Lite for Microcontrollers (TFLM). The firmware embeds StandardScaler constants and uses integer arithmetic consistent with host quantization parameters. The TFLM build includes only the operators required by the model to minimize firmware size.

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TABLE II: Model architecture (Input $L = 512$).

Layer	Configuration / Output Shape	Params
Conv2D	8 filters, kernel (64×1) → (449, 4, 8)	520
BatchNorm	–	32
MaxPool2D	(4×1) → (112, 4, 8)	0
Dropout	$p = 0.2$	0
Conv2D	16 filters, kernel (32×1) → (81, 4, 16)	4,112
BatchNorm	–	64
MaxPool2D	(2×1) → (40, 4, 16)	0
Dropout	$p = 0.2$	0
Conv2D	32 filters, kernel (16×1) → (25, 4, 32)	8,224
BatchNorm	–	128
MaxPool2D	(2×1) → (12, 4, 32)	0
Dropout	$p = 0.2$	0
GlobalAvgPool2D	→ (32)	0
Dense	32 units, ReLU	1,056
BatchNorm	–	128
Dropout	$p = 0.3$	0
Dense	16 units, ReLU	528
BatchNorm	–	64
Dropout	$p = 0.3$	0
Dense (output)	4 units, softmax	68
Total	–	**14,924**

B. Measured resource profile

After tuning the static arena and optimization of the binary size,

- Quantized model size (flash): **80 KB**.
- Firmware + TFLM + model (flash): **410 KB**.
- Peak RAM usage (TFLM arena + tensors): **120 KB**.
- Per-window inference latency (ESP32-C3, int8, $L = 512$): $\mathbf{82 \pm 7}$ ms (mean ± SD, $N = 100$).

C. Power and Energy Profiling

To evaluate the system's suitability for battery-powered edge nodes, energy consumption per inference (E_{inf}) and total operational lifespan were estimated. With the ESP32-C3 consuming approximately 25 mA at 3.3 V during active processing ($t_{inf} = 82$ ms):

$$E_{inf} = 3.3 \text{ V} \times 0.025 \text{ A} \times 0.082 \text{ s} \approx 6.77 \text{ mJ} \quad (4)$$

Given the sampling cycle of 575 ms, the device remains in a low-power idle state (approx. 1 mA) for the remaining 493 ms. The average current I_{avg} is calculated as:

$$I_{avg} = \frac{(25 \text{ mA} \times 82 \text{ ms}) + (1 \text{ mA} \times 493 \text{ ms})}{575 \text{ ms}} \approx 4.42 \text{ mA} \quad (5)$$

With a 1500 mAh battery, the node achieves a continuous operational lifespan of approximately **339** hours (≈ **14.1** days), performing over $\mathbf{2.1 \times 10^6}$ on-device inferences. This confirms the feasibility of long-term field deployment for real-time monitoring.

VII. Evaluation Protocol

A. Leakage-safe splits

Train/Validation/Test splits are **contiguous** by file to prevent any window overlap across splits. The final generalization is measured on the **12 held-out files** in `datasets/test/` (3 positions × 4 classes).

B. Metrics and Aggregation

We report window-level metrics (Precision, Recall, F1) and recording-level metrics aggregated via confidence-weighted majority voting. For each held-out recording, for each class k, we compute:

$$\text{score}_k = (\text{mean confidence}_k) \times (\text{count}_k).$$

The class with the maximum score_k is assigned.

VIII. Results

A. Window-level (Validation)

On the contiguous internal validation split (92 windows per class), window-level metrics are summarized in Table III.

TABLE III: Window-level classification metrics (validation, 92 windows per class).

Class	Precision	Recall	F1	Support
motorOFF	**1.00**	**1.00**	**1.00**	**92**
motorON	**0.98**	**1.00**	**0.99**	**92**
motorON_NoFan	**1.00**	**0.98**	**0.99**	**92**
motorON_BadFan	**0.95**	**0.93**	**0.94**	**92**
Macro avg	**0.98**	**0.98**	**0.98**	**368**

B. Recording-level (Held-out Test)

After confidence-weighted aggregation across the **12 held-out files** (3 per class), recording-level metrics are shown in Table IV. The aggregation substantially improves practical detection by smoothing transient window mispredictions.

TABLE IV: Recording-level metrics on held-out position files (12 files).

Class (files)	Precision	Recall	F1
motorOFF (3)	**0.97**	**1.00**	**0.98**
motorON (3)	**0.93**	**0.93**	**0.93**
motorON_NoFan (3)	**0.95**	**0.95**	**0.95**
motorON_BadFan (3)	**0.95**	**0.95**	**0.95**
Macro avg (12 files)		**0.95**	

C. Baseline comparisons

To contextualize the performance of the proposed method, we conducted a comparative evaluation against three representative baselines drawn from classical and deep-learning paradigms. Each baseline was implemented and tuned to provide a fair point of reference. Results are reported using the macro-averaged F1 metric.

- **Support Vector Machine (SVM):** A linear SVM trained on a set of hand-crafted time-domain descriptors (root-mean-square, kurtosis, crest factor, and shape factor). Macro F1 = **0.84**.
- **Random Forest:** An ensemble of 100 decision trees trained on the identical feature set used for the SVM,

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providing a non-linear, tree-based comparison. Macro F1 = **0.86**.

- **1D-CNN:** A conventional three-layer 1D convolutional network configured to match the proposed model's parameter count (approximately **15k** parameters), included to isolate architectural benefits. Macro F1 = **0.92**.
- **Proposed (Quantized):** The multi-channel 1D convolutional architecture with window-level aggregation, followed by post-training integer quantization targeting on-device inference. Macro F1 = $\mathbf{0.95}$.

IX. Results and Discussion

A. Performance and Resource Utilization

The proposed pipeline achieves high diagnostic accuracy within strict embedded constraints. Validation results yield a window-level macro F1-score of **0.98** (Table III), while recording-level performance on 12 held-out, position-specific files reaches **0.95** (Table IV). This confirms that confidence-weighted aggregation effectively mitigates transient window misclassifications. On-device metrics for the int8 quantized model include **80 KB Flash**, **120 KB peak RAM**, and a mean inference latency of $\mathbf{82 \pm 7}$ ms. Given the sampling stride $T_{stride} \approx 575$ ms, the system consumes only 14.2% of the available temporal budget, leaving significant headroom for auxiliary tasks or low-power sleep states.

B. Power Profiling and Error Analysis

Energy efficiency is critical for battery-powered edge nodes. With the ESP32-C3 consuming $\sim$25 mA at 3.3 V during active inference, the energy cost is approximately **6.77 mJ per inference**. Utilizing a 1500 mAh battery, the node can theoretically sustain continuous operation for **14.1 days**, performing over 2.1 million inferences. Error analysis indicates that most misclassifications occur within the *motorON_BadFan* class due to subtle vibrational similarities with normal running states. However, the orientation-invariant magnitude and elongated temporal kernels significantly enhance robustness against mounting variability and transient environmental noise.

C. Comparative Evaluation

The proposed multi-channel 1D CNN was evaluated against three baselines using the same leakage-safe protocol. The model (**0.95** F1) outperformed linear SVMs (**0.84**), Random Forests (**0.86**), and parameter-matched conventional 1D-CNNs (**0.92**). These results validate the efficacy of the magnitude-integrated feature extraction and the post-inference aggregation strategy.

D. Limitations and Industrial Outlook

Unlike many accuracy-centric studies, this work enforces **contiguous file splits** to eliminate temporal leakage and provides empirical TinyML deployment metrics. A primary limitation is the specific focus on fan-related faults in a single motor frame (0.37 kW). While fan unbalance is a critical cooling failure mode, industrial fault signatures like bearing pitting, shaft misalignment, and mechanical looseness were not included. Future research will focus on expanding the fault spectrum and evaluating cross-machine generalization using multi-modal sensor fusion. Nevertheless, the current system demonstrates a high-fidelity, sub-100 ms quantized solution suitable for immediate retrofit industrial condition monitoring.

X. Conclusion

This study presents a comprehensive TinyML framework for on-device induction motor fault detection using a low-cost MEMS accelerometer and an ESP32-C3 microcontroller. By enforcing strictly **contiguous, leakage-safe data splits**, the system ensures methodological rigor and robust generalization, achieving a window-level macro F1 of **0.98** and a recording-level F1 of **0.95**. The solution is highly resource-efficient, requiring only **80 KB** of Flash and **120 KB** of RAM. With an inference latency of **82 ms** and an estimated energy consumption of **6.77 mJ** per cycle, the system can operate continuously for approximately **14 days** on a 1500 mAh battery. This demonstrates that high-fidelity mechanical diagnostics can be localized on resource-constrained hardware with high energy-determinism. While this work focuses on fan-related anomalies, the orientation-invariant pipeline provides a scalable blueprint for broader industrial maintenance. Future research will expand the fault spectrum to include bearing defects and misalignment to further advance the versatility of Edge AI in Industry 4.0 environments.

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[0]Code and dataset available at: `https://github.com/vedantabanerjee/motorfaultdetection`

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